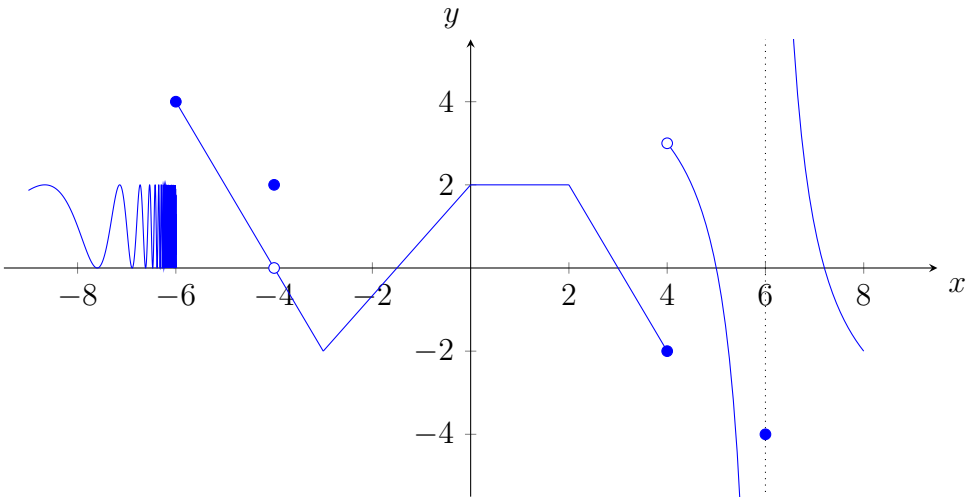
MAT148H Tutorial Quiz 2-Version A

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to -4} f(x)$

Final Answer

[redacted]

(b) $\lim_{x \to 6} \left(\frac{2}{f(x)} \right)$

Final Answer

[redacted]

(c) $\lim_{x \to -6^-} f(f(x))$

Final Answer

[redacted]

2. Let f be some function for which you know only that

$$\text{if } 0 < |x-3| < 1, \quad \text{then} \quad |f(x)-5| < 0.1.$$

Consider the implication $0 < |x-3| < 1 \implies |2f(x)-10| < C$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x\to 2} f(x) = 3$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x\to 2} \frac{x+1}{x-1} = 3$$

Do not use any limit laws or any other theorems.

[redacted]

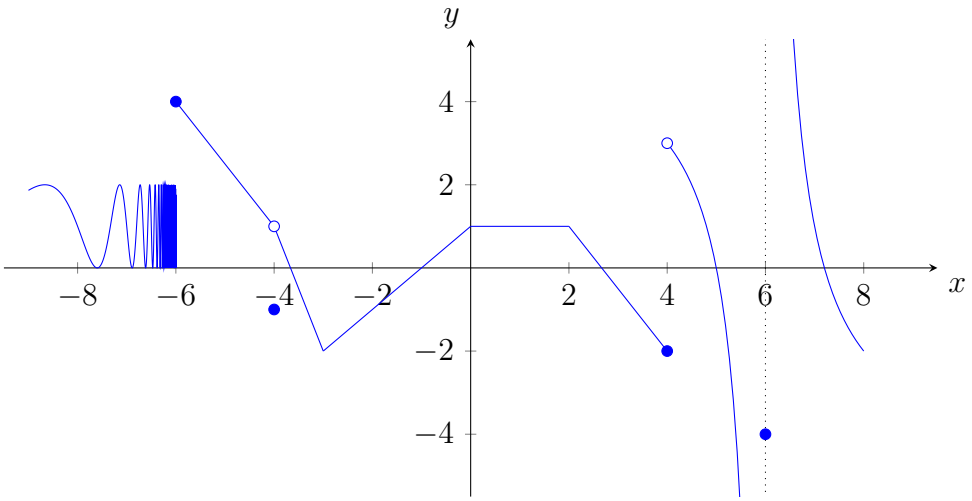
MAT148H Tutorial Quiz 2-Version B

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to -4} f(x)$

Final Answer

(b) $\lim_{x \to 6} f(x)$

Final Answer

(c) $\lim_{x \to -6^-} f(f(x))$

Final Answer

2. Let f be some function for which you know only that

$$\text{if } 0 < |x-2| < 1, \quad \text{then} \quad |f(x)-4| < 0.1.$$

Consider the implication $0 < |x-2| < 1 \implies |3f(x)-12| < C$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x\to 1} f(x) = 1$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x\to 1} \frac{2x+1}{x+2} = 1$$

Do not use any limit laws or any other theorems.

[redacted]

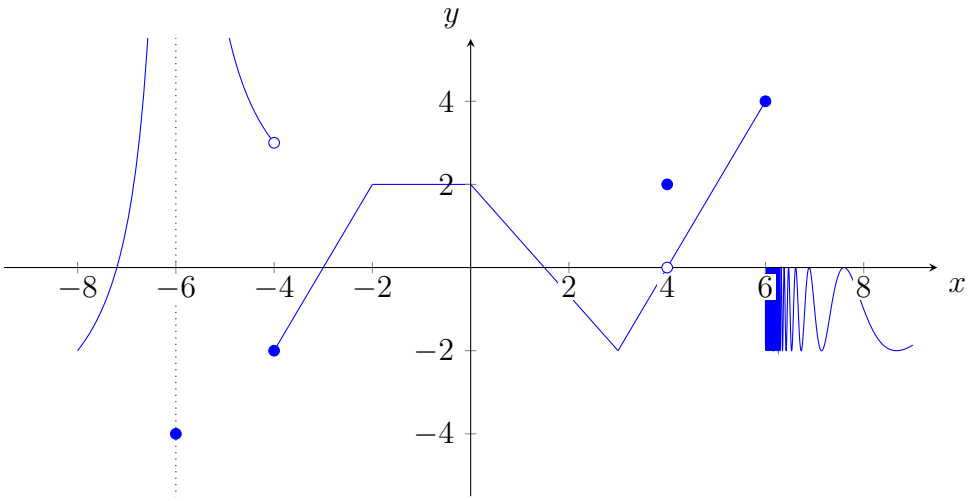
MAT148H Tutorial Quiz 2-Version C

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to 4} f(x)$

Final Answer

[redacted]

(b) $\lim_{x \to -6} f(x)$

Final Answer

[redacted]

(c) $\lim_{x \to 6^+} f(f(x))$

Final Answer

[redacted]

2. Let f be some function for which you know only that

$$\text{if } 0 < |x-1| < 1, \quad \text{then} \quad |f(x)-3| < 0.2.$$

Consider the implication $0 < |x-1| < 1 \implies |6-2f(x)| < C$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x\to 3} f(x) = 6$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x\to 3} \frac{x+3}{x-2} = 6$$

Do not use any limit laws or any other theorems.

[redacted]

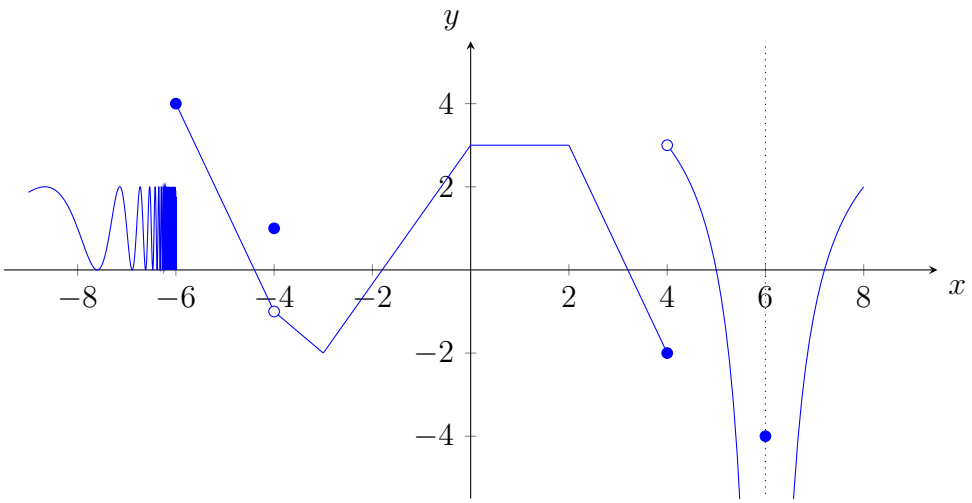
MAT148H Tutorial Quiz 2-Version D

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to 4} f(x)$

Final Answer [redacted]

(b) $\lim_{x \to 6} \left(\frac{3}{f(x)} \right)$

Final Answer [redacted]

(c) $\lim_{x \to -6^-} f(f(x))$

Final Answer [redacted]

2. Let f be some function for which you know only that

$$\text{if } 0 < |x-4| < 1, \quad \text{then} \quad |f(x)-2| < 0.1.$$

Consider the implication $0 < |x-4| < 1 \implies |4f(x)-8| < C$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x\to 2} f(x) = 1$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x\to 2} \frac{3x-1}{x+3} = 1$$

Do not use any limit laws or any other theorems.

[redacted]

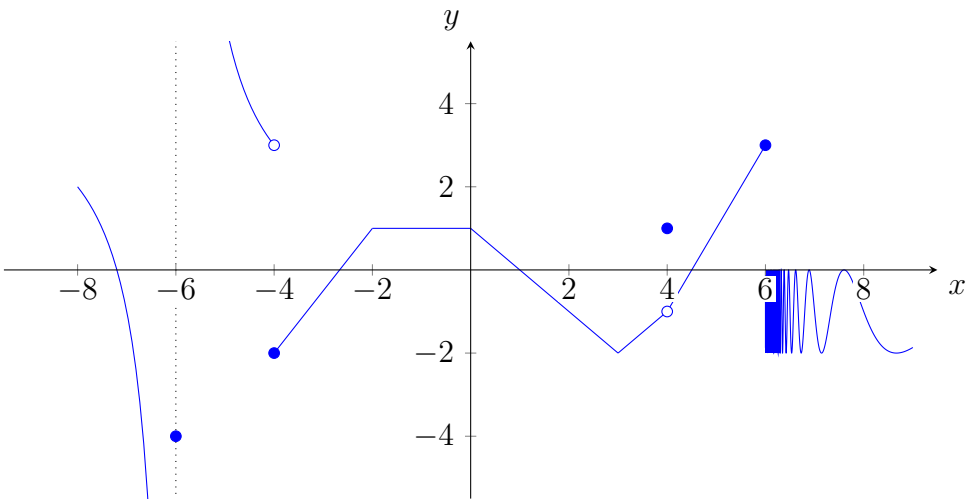
MAT148H Tutorial Quiz 2-Version E

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to -1} f(x)$

Final Answer

[redacted]

(b) $\lim_{x \to -6} \left(\frac{2}{f(x)} \right)$

Final Answer

[redacted]

(c) $\lim_{x \to 6^+} f(f(x))$

Final Answer

[redacted]

2. Let f be some function for which you know only that

$$\text{if } 0 < |x-2| < 1, \quad \text{then} \quad |f(x)-2| < 0.1.$$

Consider the implication $0 < \left|\frac{x}{2} - 1\right| < C \implies |f(x)-2| < 0.1$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x\to\infty} f(x) = 2$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x\to\infty} \frac{2x+5}{x-2} = 2$$

Do not use any limit laws or any other theorems.

[redacted]

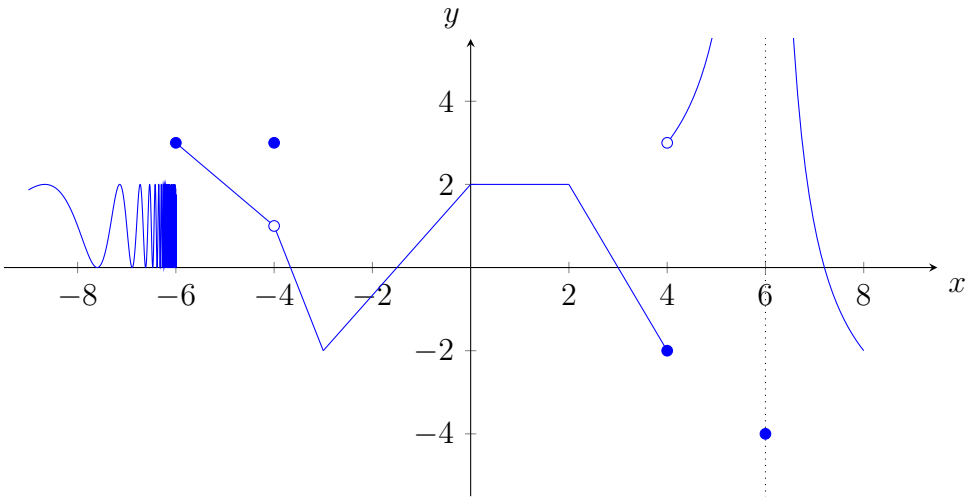
MAT148H Tutorial Quiz 2-Version F

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to -4} f(x)$

Final Answer

[redacted]

(b) $\lim_{x \to 6} f(x)$

Final Answer

[redacted]

(c) $\lim_{x \to -6^-} f(f(x))$

Final Answer

[redacted]

2. Let f be some function for which you know only that

$$\text{if } 0 < |x-3| < 1, \quad \text{then} \quad |f(x)-5| < 0.1.$$

Consider the implication $0 < |2x-6| < C \implies |f(x)-5| < 0.1$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x\to\infty} f(x) = 2$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x\to\infty} \frac{4x+1}{2x-1} = 2$$

Do not use any limit laws or any other theorems.

[redacted]

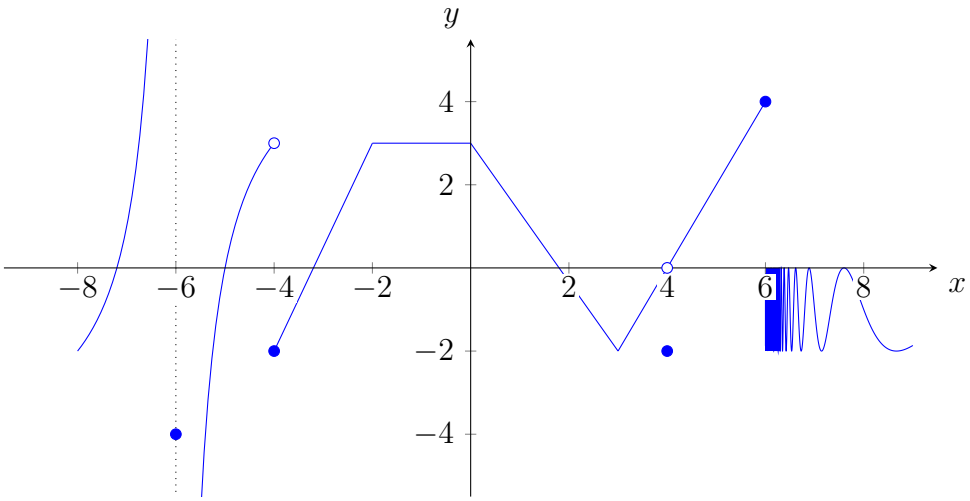
MAT148H Tutorial Quiz 2-Version G

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to -4} f(x)$

Final Answer

[redacted]

(b) $\lim_{x \to -6} \left(\frac{5}{f(x)} \right)$

Final Answer

[redacted]

(c) $\lim_{x \to 6^+} f(f(x))$

Final Answer

[redacted]

2. Let f be some function for which you know only that

$$\text{if } 0 < |x+1| < 1, \quad \text{then} \quad |f(x)-4| < 0.2.$$

Consider the implication $0 < \left|\frac{x}{3} + \frac{1}{3}\right| < C \implies |f(x)-4| < 0.2$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x\to\infty} f(x) = \frac{1}{2}$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x\to\infty} \frac{x+3}{2x+1} = \frac{1}{2}$$

Do not use any limit laws or any other theorems.

[redacted]

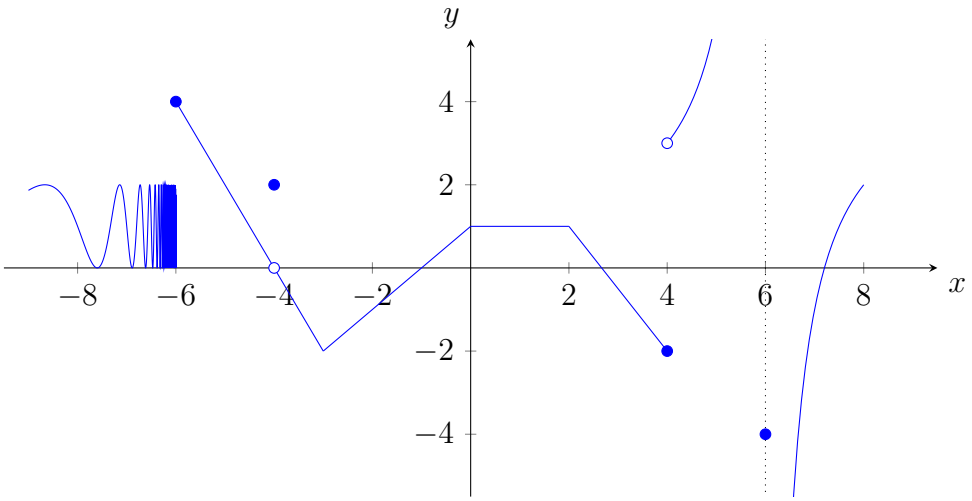
MAT148H Tutorial Quiz 2-Version H

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to 2} f(x)$

Final Answer

[redacted]

(b) $\lim_{x \to 6} f(x)$

Final Answer

[redacted]

(c) $\lim_{x \to -6^-} f(f(x))$

Final Answer

[redacted]

2. Let f be some function for which you know only that

$$\text{if } 0 < |x-4| < 1, \quad \text{then} \quad |f(x)+2| < 0.1.$$

Consider the implication $0 < |3x-12| < C \implies |f(x)+2| < 0.1$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x\to\infty} f(x) = 3$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x\to\infty} \frac{3x+2}{x-1} = 3$$

Do not use any limit laws or any other theorems.

[redacted]

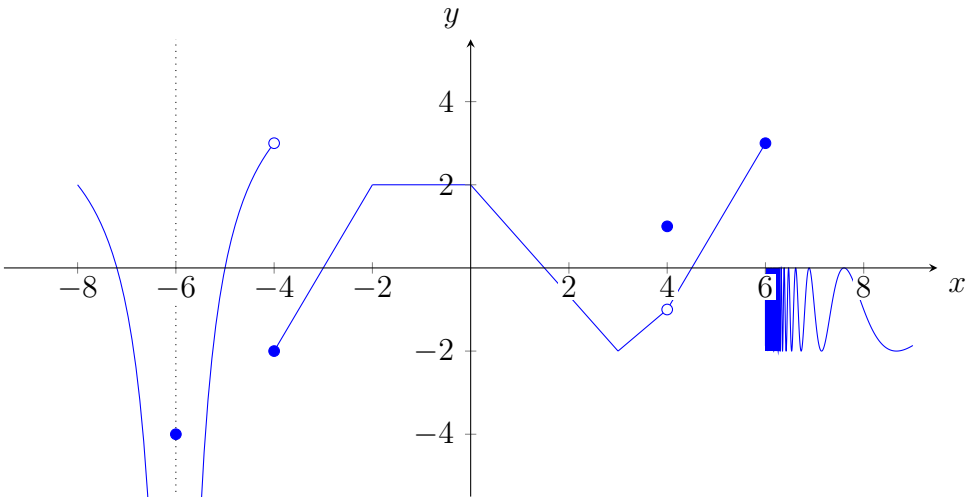
MAT148H Tutorial Quiz 2-Version I

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to -2} f(x)$

Final Answer [redacted]

[redacted]

(b) $\lim_{x \to -6} f(x)$

Final Answer [redacted]

[redacted]

(c) $\lim_{x \to 6^+} f(f(x))$

Final Answer [redacted]

[redacted]

2. Let f be some function for which you know only that

$$\text{if } 0 < |x - 6| < 1, \quad \text{then} \quad |f(x) - 1| < 0.05.$$

Consider the implication $0 < \left|\frac{x}{2} - 3\right| < C \implies |f(x) - 1| < 0.05$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x\to\infty} f(x) = 5$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x\to\infty} \frac{5x-2}{x+1} = 5$$

Do not use any limit laws or any other theorems.

[redacted]

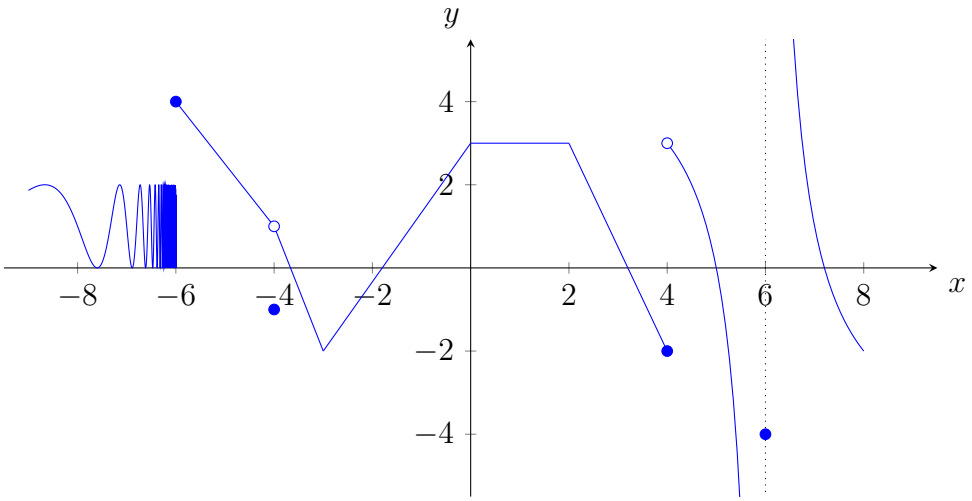
MAT148H Tutorial Quiz 2-Version J

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to -4} f(x)$

Final Answer [redacted]

(b) $\lim_{x \to 6} \left(\frac{-2}{f(x)} \right)$

Final Answer [redacted]

(c) $\lim_{x \to -6^-} f(f(x))$

Final Answer [redacted]

2. Let f be some function for which you know only that

$$\text{if } 0 < |x-2| < 1, \quad \text{then} \quad |f(x)+3| < 0.1.$$

Consider the implication $0 < |x-2| < 1 \implies |-5f(x)-15| < C$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x \to -1} f(x) = 2$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x \to -1} \frac{x+3}{x+2} = 2$$

Do not use any limit laws or any other theorems.

[redacted]

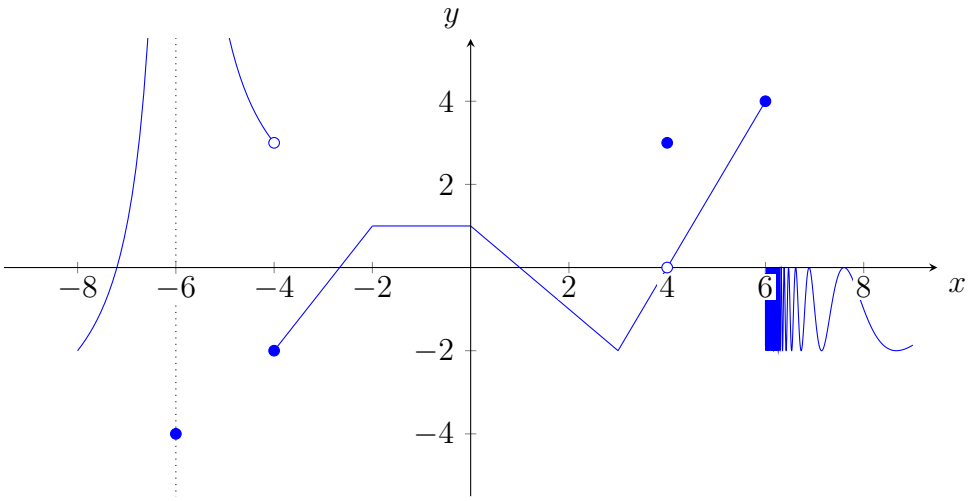
MAT148H Tutorial Quiz 2-Version K

October 13/14, 2026

Sample Solutions

For Question 1 to Question 3(a), only your final answer will be graded. **No justification is necessary.**

1. Consider the graph of $y = f(x)$ for $-9 \leq x \leq 9$.



Evaluate each limit. If a limit equals ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(a) $\lim_{x \to -4} f(x)$

Final Answer [redacted]

[redacted]

(b) $\lim_{x \to -6} f(x)$

Final Answer [redacted]

[redacted]

(c) $\lim_{x \to 6^+} f(f(x))$

Final Answer [redacted]

[redacted]

2. Let f be some function for which you know only that

$$\text{if } 0 < |x+2| < 1, \quad \text{then} \quad |f(x)-3| < 0.1.$$

Consider the implication $0 < \left|\frac{x}{4} + \frac{1}{2}\right| < C \implies |f(x)-3| < 0.1$, where C is a real number. Find *all* values of C for which this implication is necessarily true. Write your final answer by using interval notations.

Final Answer

[redacted]

3. Let $f(x)$ be a function defined on $\mathbb{R}$.

(a) Write down the formal definition of "$\lim_{x\to\infty} f(x) = 1$".

Final Answer

[redacted]

(b) Use the formal definition of the limit to prove

$$\lim_{x\to\infty} \frac{x+5}{x-3} = 1$$

Do not use any limit laws or any other theorems.

[redacted]